Sparse Coverage:

Semantic Center Representations for Patent Prior-Art Retrieval

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Patent prior-art search: a recall-driven problem

Patent claim

"An arrangement for activating and deactivating noise cancellation in a mobile station, including a detection circuit . . ."

The task. Given a query patent, retrieve earlier patents disclosing related technical ideas.

- **Term mismatch** — "*a writing instrument*" vs "*a pen*" → zero lexical overlap.
- **Recall-critical** — missing one relevant doc ⇒ legal & economic risks.

Our goal

A patent retriever that

- handles **term mismatch**
- delivers **high recall**
- stays **lightweight** (huge & growing corpus)

A naive approach: dense document representation

Patent claim

"An arrangement for activating and deactivating noise cancellation in a mobile station, including a detection circuit . . ."

encode →

Single dense vector

$\mathrm{v} \in \mathbb{R}^d$

capacity bottleneck!

component + device + circuit + function

any one may drive relevance

Where it breaks

Patent claims have combinatorially rich relevance structure $\Rightarrow$ single dense vectors are fundamentally under-sized.

On the Theoretical Limitations of Embedding-Based Retrieval

(Weller et al., ICLR 2026)

Capacity bound (Prop. 2)

$$d \;\geq\; \mathrm{rank}_{\pm}\big(2A - 1_{m \times n}\big) - 1$$

$A \in \{0,1\}^{m \times n}$: binary relevance matrix (queries $\times$ documents) • $\mathrm{rank}_{\pm} M$: **sign rank** — smallest rank of a real matrix whose entry signs match M.

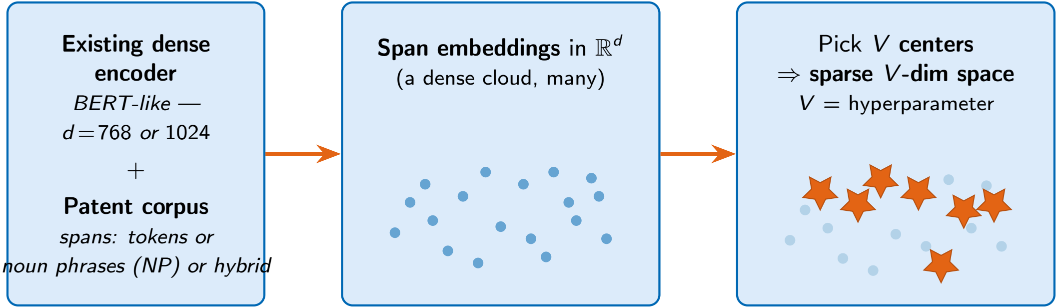
What it says

To perfectly capture the relevance structure of a task defined by A, the dense retriever's dimensionality d must be at least $\mathrm{rank}_{\pm}(2A - 1) - 1$.

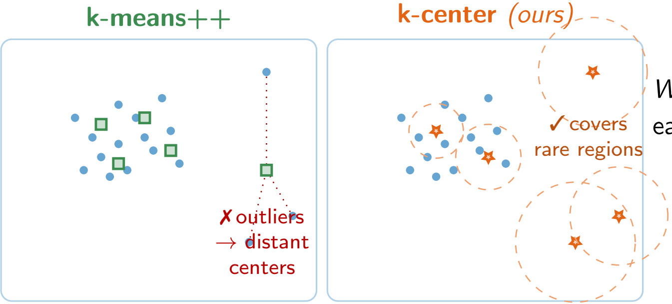
At fixed d, no training beats this bound $\rightarrow$ formalizes the failure on the previous slide.

Our idea: a tunable sparse space from a dense encoder

We turn any pretrained dense encoder into a **sparse retrieval space** of **tunable dimension** V — fully **unsupervised**.



Center selection: cover the embedding space



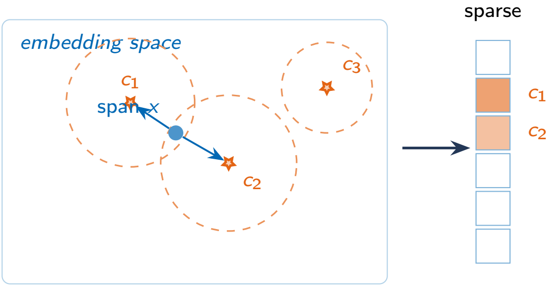
k-center objective

$$\min_{|C|=V} \max_{x \in X} \min_{c \in C} \left(1 - \langle x, c \rangle\right)$$

Worst-case coverage: pick the farthest point each step.

- **k-center** (*farthest-first traversal*, Gonzalez 1985): deterministic arg max → rare regions **always** get a center; centers are real spans.
- **k-means++**: probabilistic pick + centroid refinement → outliers **absorbed** into dense clusters.
- Per-center radius $r_c = 99^{\text{th}}$-percentile cosine distance within each Voronoi cell → adapts to local density.

From spans to sparse vectors



Each span activates the Top-K centers whose radius r_c covers it $\left(K = 5\right)$.

Document weights (max-pool over spans):

$$w_{d,c} \;=\; \max_{x_i \in S_d} \; 1[c \in A(x_i)]\, \langle x_i, c \rangle$$

Retrieval scoring

$$\mathrm{score}(q, d) = \sum_{c \,\in\, \mathrm{supp}(q) \,\cap\, \mathrm{supp}(d) \,\setminus\, C_{\mathrm{stop}}} w_{q,c}\, \widetilde{w}_{d,c}\, \mathrm{idf}_c^{\alpha}$$

- Only shared activated centers contribute.
- Standard **inverted-index traversal** — no dense scoring at retrieval time.
- Defaults: $K{=}5$, $\alpha{=}2$, stop top-1% centers.

Main results on CLEF-IP 2013

48 query topics • 17 k documents • 1.4 M passages

Model	**Unit / dim**	**V**	**R@100↑**	**mAP**	**MAP(D)**
BM25	–	–	85.90	46.62	25.91
SPLADE-v2	–	30 522	89.51	47.57	24.34
ColBERTv2	–	d=128	71.28	40.19	19.21
BERT-for-Patents	–	d=1024	66.08	33.02	19.45
PaECTER	–	d=1024	97.22	**58.61**	26.63
PatentMap-V0	–	d=1024	96.53	57.01	26.68
Ours: BERT-f-P	enc. tok.	40k	93.47 ↑27	46.05 ↑13	**30.03** ↑11
Ours: PaECTER	NP+tok.	50k	**99.31** ↑2	45.44 ↓13	25.35 ↓1
Ours: PatentMap	NP	20k	97.71 ↑1	30.90 ↓26	25.40 ↓1

↑↓ = absolute change vs. *same encoder, standalone baseline* (BERT-f-P, PaECTER, PatentMap-V0).

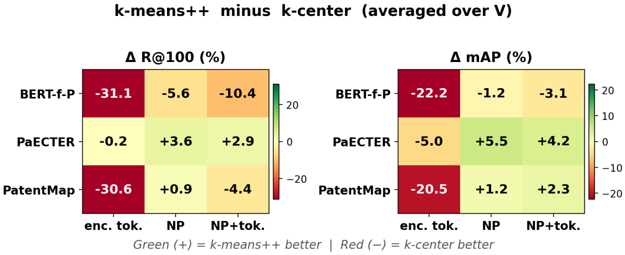
- **99.31 R@100** — highest document recall across all methods.
- **30.03 MAP(D)** — best passage ranking; >50% gain over BERT-f-P baseline.
- **PaECTER** is trained on patent **citation signals** → excels at ranking cited-relevant docs high (mAP lead). Sparse Coverage prioritizes R@100 instead.

Take-away

High-recall first-stage retriever with inverted-index efficiency.

Center selection matters: k-center vs k-means++

Only the **center-selection step** differs; encoder, spans, radii, top-K activation, and scoring stay **identical**.



Δ = k-means^{++} − k-center, averaged over V.
Red = k-center better • Green = k-means^{++} better.

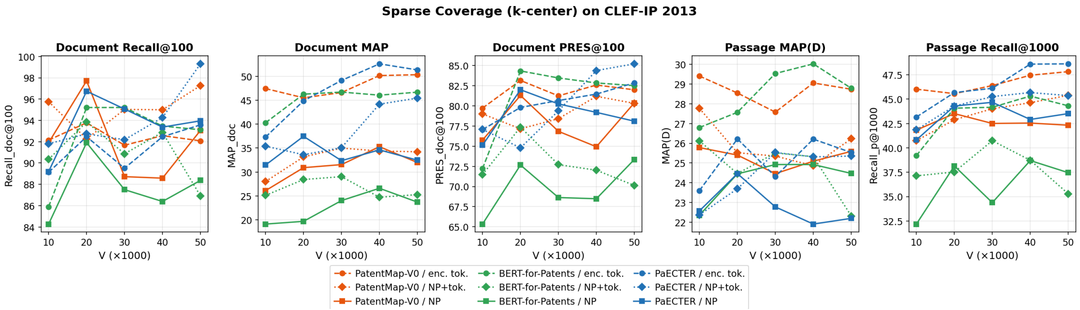
What we see:

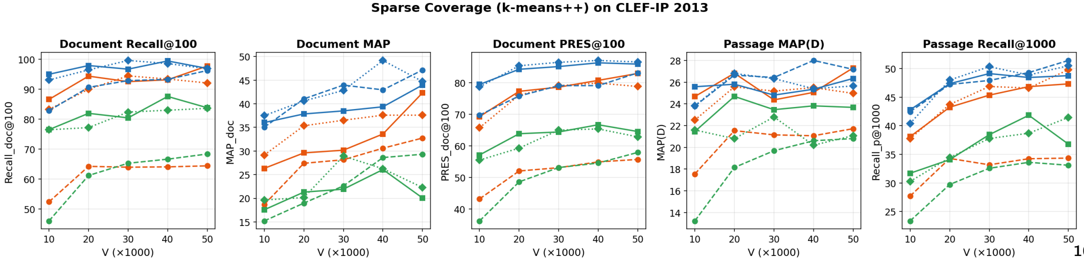
- For **encoder tokens**, k-means++ collapses (R@100 drops by 22–31 pts).
- For **NP / hybrid units**, the two strategies are close, sometimes k-means++ slightly ahead on ranking.
- **Granularity** of spans interacts with **center distribution**.

Design takeaway

Fine span units need **coverage** (k-center); coarse units tolerate **density** (k-means++).

Performance vs vocabulary size V





Conclusions

Sparse Coverage — turn any pretrained dense encoder into a sparse V-dim retriever, fully unsupervised.

- ✓ **Handles term mismatch** — semantic centers over spans capture meaning without shared lexicals.
- ✓ **High recall** — 99.31 R@100 on CLEF-IP 2013 (highest across all methods).
- ✓ **Lightweight** — inverted-index retrieval, scales to millions of patents.

`github.com/ZoeYou/patent-sparse-coverage`

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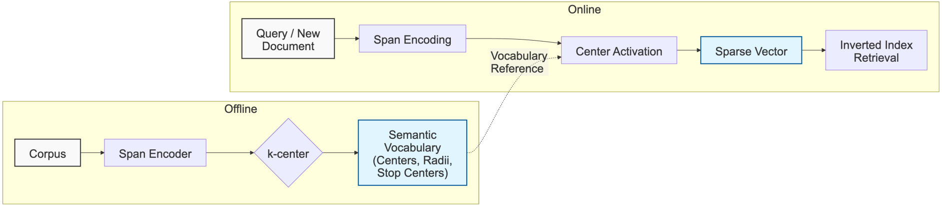
Backup: Sparse Coverage — full pipeline & data



Offline (once per corpus):
Decompose patents into spans → embed → select V **centers** that *cover* embedding space → estimate per-center radii r_c.

Online (per query / document):
Each span activates the centers whose radius covers it → **sparse vector** over centers → **inverted-index** retrieval.

Offline vocabulary corpus • EPO bulk data, English, 2000–2010 • 300 k documents stratified-sampled from ∼2 M • all sections (abstract, claims, description) → spans.

Evaluation • CLEF-IP 2013: 48 English topics, 17 k docs, 1.4 M passages.

Vocabulary construction sees no query, no qrels — no supervision leaks into centers, regardless of any document overlap.